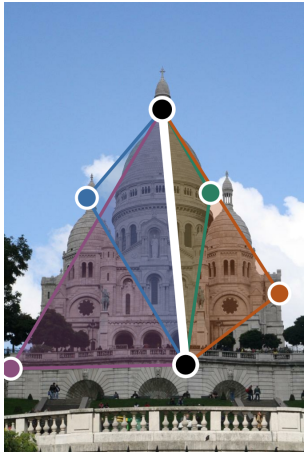
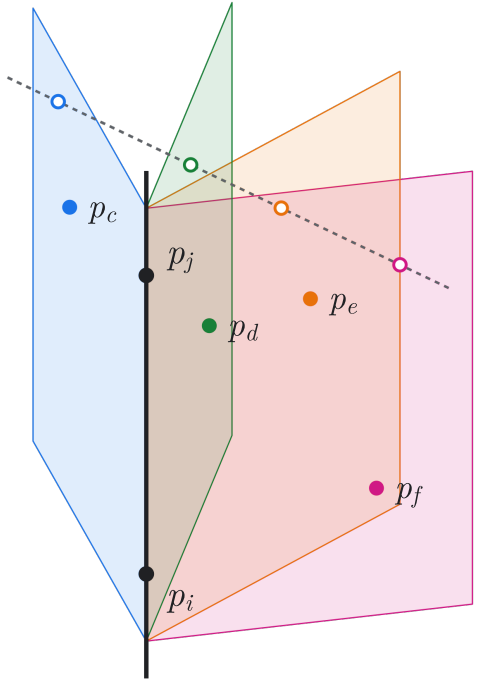
## a On a photo



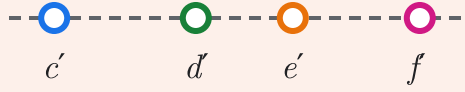
The line through $p_i,\ p_j$ (white) and four further points span four planes: a pencil. Each pair of points and 4-set of the others gives one; $n$ points give many.

## b A pencil of planes




## c One number per pencil

Any line crossing the four planes (dashed in b) meets them in four points:




$$\mathrm{CR}_{ij}(c,\ d;\ e,\ f) = \frac{(e'-c')(f-d')}{(e'-d')(f-c')} = 1.51$$

the same for every such line, and equal to $[ijce][ijdf]\ /\ ([ijcf][ijde])$.
No frame and no point scale enter.

## d Frame coordinates

Put $p_1,\ldots,p_5$ at $e_1,\ldots,e_4$, $(1,1,1,1)$ and write $p_k = (x,y,z,w)$.

$$\mathrm{CR}_{12}(3,4;\ 5,k) = z/w$$
$$\mathrm{CR}_{13}(2,4;\ 5,k) = y/w$$
$$\mathrm{CR}_{23}(1,4;\ 5,k) = x/w$$

These $3(n-5)$ pencil cross-ratios are the projective coordinates of the other points, so the family has rank $3n-15$ for every $n$.

Checked in SymPy and Macaulay2, and proved in Lean (frame_cr).